

Restricting the semantics of interrogative disjunctors

1 Introduction. In some languages, forming alternative questions (AltQs) requires a special disjunction marker (Haspelmath, 2007; Mauri, 2008). In Mandarin, AltQs are formed with the morpheme *háishì*. In this paper, I motivate a new approach to its semantics. Contra previous proposals, I argue that *háishì* is not a symmetric connective (cf. Huang, 1991; Erlewine, 2025); instead, it expresses an asymmetric mapping from a proposition p and a question Q to form a new question, roughly ‘ Q or p ?’, which is like Q , except that it also has p as a possible answer. When Q is a polar question, its positive answer q forms an AltQ together with p : “ q or p ?” To support this proposal, I observe empirical asymmetries in *háishì* constructions involving the distribution of polar question particles and the acceptability of *wh*-questions.

2 Data & previous analyses. Alternative questions in Mandarin are formed with *háishì*, as shown in (1).

- (1) Ā Máo xǐhuān hē kāfēi (ne,) { *huòzhě/ *háishì* } xǐhuān hē chá (ne)?

A Mao like drink coffee NE or or like drink tea NE

‘Which one is true: A Mao likes drinking coffee or tea?’

(AltQ)

Here *háishì* connects two seemingly declarative clauses, neither marked by any overt interrogative morpheme. We might thus assume each clause denotes a proposition, and the combination expresses an AltQ with those two propositions as its possible answers (Li and Thompson, 1981). But this analysis is challenged by the recursive uses of *háishì* as in (2); the output of one *háishì* doesn’t feed the composition with the other.

- (2) Ā Máo jīntiān (**háishì**) míngtiān **háishì** hòutiān lái (ne)?

A Mao today or tomorrow or in 2 days come NE

‘Will A Mao come today, tomorrow, or in two days?’

Huang (1991) instead has *háishì* express the Boolean operator $\vee$ and derives the AltQ by scoping the disjunction over an interrogative complementizer (after Karttunen, 1977; von Stechow and Heim, 2021). More recently, Erlewine (2025) has proposed instead that *háishì* expresses set union in the domain of Focus Semantic Values (Rooth, 1992; Beck, 2006). Under this approach, the denotation of *háishì* is compatible with disjuncts of varying semantic types, as in (3). The latter analyses both allow the recursive uses of *háishì*.

- (3) $\llbracket \beta \text{ or } \textit{háishì} \alpha \rrbracket^{alt} = \llbracket \beta \rrbracket^{alt} \cup \llbracket \alpha \rrbracket^{alt}$ (4) **LF:** $[\beta [\textit{háishì} \alpha]]$

For simplicity I assume that *háishì* takes just two arguments with the LF in syntax in (4), where *háishì* merges first with the right clause α and then the left clause β , though see Erlewine (2025) for required elaborations.

Semantically, all of the previous analyses assume that disjuncts connected by *háishì* are symmetric. Both Li and Thompson (1981) and Huang (1991) require that both disjuncts are of a proposition type, and Erlewine (2025) allows *háishì* to apply to any arguments, so long as both have denotations of the same semantic type.

This paper challenges such symmetry, with evidence from question particles and *wh*-questions. I propose that *háishì* is a type $\langle st, \langle stt, stt \rangle \rangle$ operator, composing first with a proposition, and then with a question.

3 Challenge 1: question particles. As in (5), the right argument of *háishì* cannot be a *ma*-marked clause. Such a restriction does not apply to the left clause—it can be either *ne*-marked or *ma*-marked (6).

- (5) *Ā Máo xǐhuān hē kāfēi { **ma**/ *ne*/ $\emptyset$ }, *háishì* xǐhuān hē chá **ma**? *[… *háishì* … *ma*]

A Mao like drink coffee QP NE or like drink tea QP

(Judgment from Mainland Mandarin speakers; cf. Taiwan Mandarin, Hsieh 2026)

- (6) Ā Máo xǐhuān hē kāfēi { **ma**/ *ne*/ $\emptyset$ }, *háishì* xǐhuān hē chá (ne)? ✓[… *háishì* … (*ne*)]

A Mao like drink coffee QP NE or like drink tea NE

‘Does A Mao like drinking coffee, or does he like drinking tea?’

The particle *ma* occurs only with matrix PolQs but not other kinds of questions (cf. (7)&(8)). It’s thus commonly understood as marking a PolQ (Cheng, 1991; Paul, 2014; Pan, 2019, a.o.). This rules out the possibility that *ma* scopes above *háishì* yielding an AltQ reading. But since the left clause can be *ma*-marked, it’d be plausible for *ma* in the right clause to scope under *háishì*. If so, I suggest that the contrast ((5) vs. (6)) can be explained if the right argument of *háishì* **cannot** be a PolQ—what it is in (6) is instead a proposition.

- (7) Ā Máo xǐhuān hē kāfēi **ma**?

A Mao like drink coffee QP

‘Does A Mao like drinking coffee?’

- (8) Ā Máo xǐhuān hē shénme **ma**?

A Mao like drink what QP

#‘What does A Mao like drinking?’

4 Challenge 2: *wh*-questions. Let’s now take an uncontroversial question and examine when it can occur in a *háishì*-construction. As in (9), using *háishì* for a sequence of two *wh*-questions is bad. However, it becomes acceptable if the right argument is arguably not a question, as in (10). We thus observe another asymmetry.

- (9) Nǐ de diànhuà shì **duōshǎo**, {huòzhě /*háishì} nǐ de yóuxiāng shì **shénme**? [*whQ+whQ*]
 2SG DE phone number be how many or or you DE email be what
 ‘(To recover the username, please tell me) what is your phone number, or what is your email?’
- (10) Nǐ dǎsuàn zài **nǎr** zū chē (ne), háishì dǎsuàn jiè yí-liàng (ne)? [*whQ + háishì(p)*]
 2SG plan LOC **where** rent car NE or plan borrow one-CL NE
 ‘Where do you plan to rent a car, or do you plan to borrow one?’ (after Ciardelli et al., 2018)

If, as suggested by Erlewine (2025), the arguments of *háishì* are focus alternatives, where propositions and questions share the semantic type in their Focus Values, we are left with no semantic explanation for why (5) and (9) are ungrammatical with *háishì* (cf. (6) and (10)). Similarly, if *háishì* expresses a Boolean $\vee$ (Huang, 1991), following Karttunen (1977) and von Stechow (2011), a logical connective *háishì* would make available an LF like (11) for the *wh*-question disjunction in (9), where two disjuncts are of a proposition type, and the question reading is derived via ATB-movement of Op_{st} . However, the sentence is ungrammatical.

- (11) *LF: [Op_1 [*what*₃ [C_Q t_1] your-number-is t_3] *háishì*_(t,t,t) [*what*₄ [C_Q t_1] your-email-is t_4]]]

5 Proposal: [Q [*háishì* p]]. I consider the observed semantic asymmetries to be reflected in the semantics of *háishì*: its right argument must be a proposition, and the left one in fact *must* be a question: a PolQ as in (6), a *wh*-Q as in (10), and even in cases without interrogative morphology, such as (1)—I hypothesize that because, *modulo* prosody, a non-*ma*-marked clause like *Ā Máo xǐhuān hē kāfēi* (cf. (7)) is ambiguous between the declarative and PolQ readings. Together with (4), I posit the lexical entry for *háishì* in (12), where it takes a proposition p and adds it to a question Q to form a new question. For PolQs, I assume a monopolar analysis in which Q denotes a singleton set (Roberts, 1996; Uegaki, 2014). I leave the source of the familiar exclusivity inference of AltQs independent from the type restriction proposed here.

- (12) $\llbracket háishì \rrbracket^w := \lambda p_{st}. \lambda Q_{stt}. \lambda r_{st}. [Q(r) \vee r = p]$ ((12) $\equiv \lambda p. \lambda Q. [Q \cup \{p\}]$)

One might wonder if *háishì* does *not* scope over the preceding questions in (6) and (10), and they both involve two separate questions. I leave it open and maintain that to saturate the second argument of *háishì*, one can assume that the left argument is a covert pronoun anaphoric to a salient question. For example, in (10), the salient question would be the preceding *wh*-question, and the right argument of *háishì* would be an answer to the output of *háishì*: (i) the question (10) is not a PolQ canceling the previous *wh*-question, as polar answers are infelicitous; (ii) asserting α entails the negation of all the possible answers of the *wh*-question, as in (13).

- (13) Wǒ dǎsuàn jiè Ā Máo de chē.

1SG plan borrow A Mao DE car

‘I plan to borrow A Mao’s car.’

$\Rightarrow$ *I plan to borrow a car.* $\rightsquigarrow$ *I don’t plan to rent a car.*

6 Discussions. (a) Rightward is upward. To account for the recursion fact in (2), one might have to assume the structure for (2) to be (14) instead of (15), where the rightward *háishì* is higher in the structure, because in (15) the boxed part outputs a question and bleeds the subsequent composition. One potential support comes from (16), which shows that the immediate right clause of each *háishì* has to be a proposition. Assuming (14) allows us to capture the semantic asymmetry in both *háishì*s. Were (15) assumed, since β is not an argument of the outer *háishì*, we would expect no semantic restriction imposed on β , contrary to the fact.

- (14) $\llbracket [\gamma [\text{háishì } \beta]] [\text{háishì } \alpha]] \rrbracket$ (15) $\llbracket [\gamma [\text{háishì } [\beta [\text{háishì } \alpha]]]] \rrbracket$

- (16) a. *(5), or does he like juice (ne)? [**ma...ma...ne*] b. (6), or does he like juice (ne)? [*✓ma...ne...ne*]

(b) Island-insensitivity. The current proposal builds AltQs without movement, which also predicts that the *háishì* disjunction in AltQs is island-insensitive (Huang, 1991; cf. Erlewine, 2025). **(c) Remaining issues.** For clause-medial *háishì*-AltQs in Mandarin, I have to assume they all involve ellipsis from biclausals. Another disjunct, *huòzhě*, never forms AltQs in Mandarin (1), suggesting that its semantics is also restrictive enough to block these constructions altogether. We also call for revisiting cross-linguistic interrogative disjunctors (e.g., Basque, Finnish, etc.) to see if they instantiate similar constraints on disjoining questions.

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